

MΛNDOROS
Meta-Analysis Network for Data, Outcomes & Research Optimization Synthesis
— From Raw Trials to Global Clinical Verdicts —
PRISMA 2020 Compliant Cochrane Handbook Core wviechtb/metafor-Grade REML Graph-Theoretic NMA & SUCRA In-Silico Virtual Trial Extrapolation
Flagship Academic Keynote & Platform Specification Dossier
Presented for Global Universities, Clinical Trial Principal Investigators & Systematic Reviewers
Live Platform: **https://mandoros.onrender.com**
MANDOROS Biostatistical Evidence Intelligence SLIDE 01 / 12
Overview

Agenda & Table of Contents

MANDOROS KEYNOTE
PART I · FOUNDATIONS

1. The Systematic Review Bottleneck

How unstructured clinical literature creates a 2-year lag in evidence adoption, and how MANDOROS solves it.

2. System Architecture & Engineering

The deterministic code stack: Python 3.10, metafor biostatistics, LLM reasoning, and zero-imputation guardrails.

PART II · ENGINES

3. Stage-1 Atomic Curation

7 relational sheets, 88 canonical fields, 4 relational eligibility operators, and verbatim page/quote provenance.

4. Phase-2 Meta-Analysis & SUCRA

Restricted Maximum Likelihood (REML), Knapp-Hartung correction, indirect comparisons, and Monte Carlo SUCRA rankings.

PART III · DISCOVERY & PRACTICAL

5. In-Silico Synthesis & Virtual Trials

Cross-trial surrogate covariance imputation, missing endpoint reconstruction, and optimal next-gen trial protocol design.

6. University User Guide & BYOK Tutorial

Step-by-step user walkthrough and illustrated tutorial on connecting your own free Google AI Studio API key.

Problem & Solution

The Systematic Review Crisis vs The MANDOROS Solution

EVIDENCE BOTTLENECK

THE PROBLEM

The Evidence Bottleneck in Modern Medicine

- **Manual Review Latency:** Traditional Cochrane/PRISMA systematic reviews require 12 to 24 months of manual human curation.
- **Unstructured PDF Silos:** Critical trial endpoints, standard deviations, and eligibility criteria are trapped in multi-column PDF tables and appendices.
- **Aggregation Bias & Inconsistency:** Trials employ divergent outcome scales, different follow-up timepoints, and varying dosage schedules.
- **LLM Hallucination Risk:** Standard AI chatbots invent sample sizes, confabulate p-values, and hallucinate treatment effects when given complex RCTs.

THE MANDOROS SOLUTION

Zero-Hallucination Deterministic Synthesis

- **Seconds, Not Years:** Converts 50-page complex clinical PDFs into standardized 7-sheet relational databases in under 45 seconds.
- **Verbatim Page & Quote Coordinates:** Every single number is cryptographically bound to exact manuscript text coordinates.

- **Pure metafor Biostatistics:** Python-implemented Restricted Maximum Likelihood (REML) and Knapp-Hartung variance estimators.
- **Collective In-Silico Intelligence:** Goes beyond meta-analysis to reconstruct unmeasured endpoints and design the next breakthrough trial.

From Raw Literature to Verified Clinical Intelligence SLIDE 03 / 12
Engineering

System Architecture & How MANDOROS Was Built

UNDER THE HOOD

LAYER 1 · INGESTION

Zero-Loss PDF Deconstruction

PyMuPDF / pdfplumber pipelines extract high-fidelity layout blocks, tables, footnotes, and vector graphics without optical degradation or OCR truncation.

LAYER 2 · EXTRACTION

Dual-Layer Consensus AI

Guided schema prompts enforce Pydantic AST validation, deterministic value/operator splits, and explicit `not_reported` epistemic boundaries.

LAYER 3 · BIOSTATISTICS

metafor & netmeta Engine

Replicates Wolfgang Viechtbauer's R `metafor` package in Python: REML, DerSimonian-Laird, Sidik-Jonkman, Knapp-Hartung, and Bucher NMA.

LAYER 4 · INTERFACE

Apple Glass Bento UI

Built on Gradio 6 / FastAPI with an Obsidian glassmorphism design system, responsive tabs, multi-file drag-and-drop studio, and interactive Plotly visuals.

🔒 **Epistemic Guardrail Law: Absolute Anti-Hallucination Guarantee** Deterministic Verification

If an endpoint, standard deviation, or patient count is omitted from the original published trial, MANDOROS strictly outputs `not_reported` during Stage-1 extraction. The platform never hallucinates baseline metrics or clinical values.

Engineered for Academic Rigor & Reproducible Biostatistics SLIDE 04 / 12
Pipeline

The 8-Node Process Architecture (M-A-N-D-O-R-O-S)

ACRONYM BLUEPRINT

M

Meta-Analysis

Inverse-variance random-effects pooling, DerSimonian-Laird, REML & Knapp-Hartung variance estimation.

A

AI & Automation

Zero-loss clinical parser binding every extracted number to exact manuscript page & verbatim quote coordinates.

N

Network (NMA)

Bucher indirect treatment comparisons and Monte Carlo SUCRA cumulative surface rankings across competing drugs.

D

Data Schema

88 canonical relational columns deconstructing trial criteria into deterministic value, operator, and unit fields.

O

Outcomes

Granular tracking of continuous efficacy, binary adverse events, time-to-event survival, and longitudinal curves.

R

Research Rigor

Full alignment with PRISMA 2020, CONSORT 2010 checklists, Cochrane Risk of Bias, and GRADE framework.

O

Optimization

Scale-free Standardized Mean Difference (Hedges' g SMD) clinical domain bridging across heterogeneous rating scales.

S

Synthesis

Cross-trial surrogate covariance imputation, dose-duration meta-regression, and de-novo virtual clinical trial synthesis.

Stage-1 Atomic Curation: 7 Relational Sheets & 88 Fields

SINGLE-STUDY EXTRACTION

Relational Sheet	Target Clinical Entity	Key Atomic Fields Extracted	Verification & Provenance Standard
1. Master_Extraction_Sheet	Executive Overview	Study ID, Title, DOI, Phase, Total N, Primary Endpoint, Safety Verdict	Exact page number & verbatim sentence coordinates
2. Study_Characteristics	Methodological Rigor	Allocation, Masking, Design Type, Centers, Funding, Risk of Bias	CONSORT Items 1–10 checklist audit
3. Population_Baseline	Patient Demographics	Mean/SD Age, % Female, BMI, Disease Duration, Baseline Diagnostic Markers	Cross-arm arithmetic conservation checking
4. Intervention_Arms	Pharmacological Regimens	Arm Name, Active Ingredient, Dosage, Route, Frequency, Titration Schedule	Standardized to WHO ATC & RxNorm vocabulary
5. Outcomes	Granular Endpoints	Timepoint (Wks), Metric, Central Tendency, Dispersion, p-value	Deterministic Value / Operator / Unit separation
6. Variables_Matrix	Trajectory Profiling	Normalized baseline vs post-baseline delta trajectories across all treatment arms	Scale normalization & change-from-baseline auditing
7. Eligibility_Criteria	Inclusion / Exclusion	Criterion Category, Atomic Name, Relational Operator, Threshold, Unit	4-Operator mathematical validation (<, ≤, >, ≥)

Phase-2 Multi-Trial Evidence Synthesis & metafor Biostatistics

EVIDENCE POOLING
RANDOM-EFFECTS POOLING

Restricted Maximum Likelihood (REML)

Executes DerSimonian-Laird and REML models with Knapp-Hartung variance adjustment. Automatically calculates:

- **Heterogeneity Metrics:** Cochran's Q, between-study variance $ au^2$, and Higgins' I^2.
- **Confidence & Prediction Intervals:** 95% Confidence Interval (mean effect) AND 95% Prediction Interval (PI for future trials).
- **Small-Study Bias:** Egger's linear regression test for funnel plot asymmetry and Begg's rank test.
- **Study Influence:** Leave-one-out sensitivity analysis detecting outlier-driven significance.

NETWORK COMPARISONS & RANKING

Graph-Theoretic NMA & SUCRA

When pooling trials that evaluated different active ingredients or comparator doses, MANDOROS unlocks:

- **Bucher Indirect Comparisons:** Indirect relative risk calculation across connected multi-trial networks.
- **SUCRA Probability Surfaces:** Surface Under the Cumulative Ranking curves via Monte Carlo simulations to rank treatments from best to worst.
- **Scale-Free SMD Domain Bridging:** Hedges' g standardization harmonizing disparate clinical questionnaires into unified domains.
- **Subgroup Heterogeneity:** Cochran's $Q_{ ext{between}}$ partitioning variance across dosage tiers and follow-up lengths.

In-Silico Emergent Synthesis: Inventing the Next Breakthrough Trial

IN-SILICO DISCOVERY
SURROGATE IMPUTATION

Missing-Data Matrix Reconstruction

If Trial A measured Endpoint X and Y at Week 24, while Trial B only measured X, MANDOROS uses cross-trial surrogate covariance and longitudinal spline interpolation to scientifically infer unmeasured trial cells with 95% uncertainty bounds.

PREDICTIVE SIMULATION

Dose–Duration Virtual Trial Simulator

Researchers can drag dosage (0–3,000 mg/day) and duration (4–156 weeks) sliders to forecast therapeutic responses with automated Extrapolation Risk Ratings (Safe Interpolation vs Mild vs Extreme Extrapolation).

DE-NOVO DESIGN

Synthesized Trial Protocol Blueprint

Synthesizes the optimal next-generation trial: pinpointing the therapeutic sweet-spot dose before toxicity escalates, trajectory plateau week, super-responder phenotype, and statistical sample size for 90% power.

From Retrospective Meta-Analysis to Prospective Trial Synthesis SLIDE 08 / 12
User Manual

Step-by-Step User Walkthrough: How to Use MANDOROS

PRACTICAL GUIDE
1
Stage 1: Extract Single Published RCT (PDF)
Open Tab 1 ("Single-Study Deep Extractor"), drop any clinical trial PDF protocol or manuscript, select your AI model, and click "Run Academic Deep Extraction". Inspect the 7 relational sheets instantly.
2
Stage 2: Ingest Multi-Trial Cohort
Switch to Tab 2 ("Phase 2 Multi-Trial Synthesis"). Choose an existing folder in `analysis/` OR select "Upload Multiple Files" and drop multiple PDFs/Excels using the glowing batch uploader.
3
Stage 3: Inspect Divergence & REML Pooling
Click "Run Multi-Trial Meta-Analysis". View the Executive Verdict card, the Divergence Adaptation Strategy, interactive Forest/Funnel plots, SUCRA rankings, and Cochran Q between-subgroup statistics.
4
Stage 4: Simulate Virtual Trials & Export Workbooks
Simulate custom dose-duration regimens, review the Reconstructed Missing Data Table, and download the 10-sheet Master Cohort Excel, interactive HTML dashboard, and ZIP archive.
Intuitive Interface Designed for Researchers, Students & Clinicians SLIDE 09 / 12
Tutorial

How to Connect Your Google AI Studio API Key (BYOK Guide)

FREE API SETUP

> 💡 **Why Connect Your Own API Key? (Bring Your Own Key — BYOK)**
>
> Connecting your personal Google AI Studio key gives you **unlimited free quota**, ultra-fast Gemini 2.0 Flash / 1.5 Pro processing speeds, prevents shared server rate-limits, and ensures your clinical research manuscripts are processed within your private Google Cloud quota!

01

Visit Google AI Studio

Go to `aistudio.google.com/app/apikey` in your browser and sign in with your regular Google account.

02

Create Free API Key

Click the blue button labeled **"Create API key"**. Select any existing Google Cloud project or click "Create API key in new project".

03

Copy Your Key

Copy the generated key string (which begins with `AIzaSy...`). It is 100% free with generous daily rate limits.

04

Paste into MANDOROS

Paste the key into the **"AI Provider API Key"** password box in MANDOROS. The platform automatically unlocks deep clinical reasoning.

100% Free · Private Data Processing · Maximum Speed SLIDE 10 / 12
Compliance

Universal Export Center & Institutional Governance

ACADEMIC STANDARDS
PUBLICATION-GRADE ARTIFACTS

Instant Export Formats

- **7-Sheet Single-Trial Excel:** Master extraction, characteristics, baseline, arms, outcomes, matrix & eligibility.
- **10-Sheet Master Cohort Workbook:** Pooled comparison matrix, heatmap, meta-analysis estimates, SUCRA rankings, and virtual trial blueprint.
- **Standalone Interactive HTML Dashboard:** Embedded Forest plots, Funnel plots, and SUCRA curves ready for publication and peer-review.
- **Structured JSON Protocol Trees:** Complete programmatic API schema for computational biology pipelines.

REGULATORY COMPLIANCE

International Scientific Standards

- **PRISMA 2020:** Conforms to Preferred Reporting Items for Systematic Reviews and Meta-Analyses.
- **CONSORT 2010:** Aligned with Consolidated Standards of Reporting Trials guidelines.
- **Cochrane Handbook Core:** Standardized effect size estimators, inverse-variance weighting, and risk-of-bias stratification.
- **GRADE Framework Ready:** Explicit confidence grading for all reconstructed and pooled evidence recommendations.

Ready for Harvard, Oxford, Nature Medicine & The Lancet SLIDE 11 / 12
Conclusion

Conclusion & Global Academic Vision

THE FUTURE OF EVIDENCE



Accelerating Medical Discovery from Raw Trials to Global Verdicts

MANDOROS bridges the gap between massive clinical trial literature and rapid evidence-based medicine. By eliminating manual transcription, enforcing zero-hallucination verbatim citations, and deploying metafor-grade biostatistics, we empower academic institutions, healthcare systems, and pharmaceutical innovators to discover insights that save lives.

Explore Live: **https://mandoros.onrender.com** · **GitHub:** **github.com/ahmedmando133/MANDOROS**